Systematic Review

State-of-the-Art Research Survey in Deep Learning

Kenji Omar Ramírez[*1], Alejandro Elí Acosta Flores[†1], Gabriel Emmanuel Monroy Romo[†1] and María de los Angeles Rabelero Campos[†1]

[1]*Master in Applied Artificial Intelligence, Instituto Tecnologico y de Estudios Superiores de Monterrey*

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Abstract

The rapid evolution of deep-learning, particularly through diffusion models and generative architectures, has transformed the audiovisual landscape. This paper surveys the current state of image and video generation, focusing on their integration into artistic workflows for cinema and video games. We analyze the technical acceleration provided by these models, such as Text-to-3D and physics-grounded motion, while critically addressing the ethical crises they precipitate. Central to this discussion are the challenges of copyright, regional stereotypes, and the rise of deepfakes. By reviewing recent literature (2024-2026), we propose a framework that balances technical efficiency with ethical responsibility and human-centric authorship.

Keywords: *AI ethics, Misinformation Risks, Intellectual Property (IP) Concerns, Responsible AI, Copyright compliance*

1. INTRODUCTION

In the current technological epoch,Artificial Intelligence (AI) has transcended its role as a mere automation tool to become a "collaborative agent". In the audiovisual and video game industries, generative models do not merely replace the artist but establish an interactive dialogue that optimizes creative phases previously burdened by technical repetition [1].

However, this democratization of high-fidelity content generation has blurred the lines between reality and simulation, sparking a profound crisis of authenticity and legal ownership. This review explores the technical breakthroughs in diffusion models and the `socio-ethical` friction they generate within the artistic community.

Note

Socio-ethical refers to the combined social consequences and ethical obligations of a technology.

$$Socio\text{-}ethical = socialimpact \bullet ethicalduty \tag{1}$$

2. LITERATURE REVIEW

2.1. Technical Acceleration and the "Collaborative Agent" Paradigm

The adoption of AI into the artistic workflow has shifted the focus from manual execution to conceptual direction. Recent work on diffusion-based visual art creations has identified a cluster of research around generation with emphasis on guided outputs (controllability), pipeline-ready assets (application-oriented), and historical or genre-specific art creation, that are particularly relevant for film and video games pipelines [8].

In this context, AI serves as a tool that suggests and transforms, allowing artists to address complex tasks such as lighting adjustments or composition cleaning in minutes [1]. This efficiency extends to the generation of 3D assets. Wang and Fu demonstrate that integrating Large Language Models (LLMs) with Gaussian Splatting (GS) enables the creation of `high-fidelity` 3D objects that exhibit `physics-grounded` motion [9].

These advancements are crucial for the video game industry, where the ability to generate `physics-aware` objects from text prompts dramatically reduces the gap between conceptualization and real-time implementation.

Furthermore, this technological leap is democratizing the video game industry. By reducing reliance on large art departments, independent teams can now achieve a level of visual and technical polish once reserved only for AAA studios with multi-million dollar budgets [2].

This shift has enabled small independent teams to compete in terms of `high-fidelity` production, refocusing the competitive advantage from sheer labor capacity to narrative and mechanical innovation. Conversely, for AAA studios this acceleration facilitates a transition toward `procedural oversight`, where the challenge lies not in asset volume but in maintaining a unique artistic signature within an increasingly automated production ecosystem.

2.2. The Crisis of Attribution and Originality

Despite these technical gains, the `AI-ification` of art has sparked backlash and rejection amid the perception of AI as "institutionalized plagiarism". Panera et al. argue that since models are trained on a vast corpora of human work without explicit consent leads the artistic community to regard the outputs as devaluing human labor [5].

[*]Corresponding authors: A01796851@tec.mx (K. O. Ramírez), A01100001@tec.mx (A. E. Acosta), A01797342@tec.mx (G. E. Monroy), A01796851@tec.mx (M. A. Rabelero)
[†]These authors contributed equally to this work.

Note

By **AI-ification of art** we refer to the systematic integration of AI models throughout the creative pipeline. Work traditionally performed manually by an artist is increasingly executed by algorithms while the artist provide conceptual direction, constraints, and final judgment.

This critique should be more precisely framed as a claim of labour theft rather than plagiarism alone. As Goetze et al. argue, the central ethical issue is not merely misattribution but the appropriation of the products of creative labour without consent nor compensation [3].

Critically, although defenders argue that AI mimics the human neural processes of learning from influence, the scale and speed of appropriation introduce a qualitative difference. The core question of the debate is whether a `prompt` constitutes sufficient creative intent to grant authorship, or whether the lack of human emotional experience in the generative process inherently devalues the art [5].

To counter this `black-box` opacity, Sankaran et al. introduce the Compositional Provenance Framework (CPF), a `socio-technical architecture` designed to restore forensic transparency across the generative pipeline. They argue that the current attribution crisis arises when models replicate stylistic signatures without a traceable record of the creative lineage. By implementing gradient-based influence tracking and immutable provenance store, CPF quantifies the training-data contributions at the token or pixel level [6].

Note

A **socio-technical architecture** is a system with human controls (roles, policies, audit and dispute workflows) and policies (content credentials, watermarking, perceptual hashing, immutable ledgers) use for tracking tech, yielding verifiable and enforceable transparency across the generative pipeline.

This technical shift is fundamental for the industry as it moves the conversation from an irreconcilable ethical deadlock toward a measurable system, where creative borrowing becomes visible and contestable,potentially paving the way for automated royalty systems and the protection of artists moral rights in the age of synthetic media.

2.3. Socio-Technical Biases and Regional Stereotypes

A significant shortcoming in the current generative models is the persistence of `regional stereotypes`. Jha et al. introduce the `ViSAGe framework`, revealing that text-to-image models often default to stereotypical markers, particularly in the depictions of Global South identity groups [4].

Note

A **ViSAGe** is a bias-auditing framework for text-to-image models that stress-tests neutral prompts across regions and identities, quantifying stereotype leakage via controlled generations, attribute tagging, and diversity/fairness metrics.

This phenomenon persists even when prompts are explicitly neutral. For the video game industry, which strives for global representation, this inherent bias in deep-learning architectures poses a risk of reinforcing harmful tropes through automated content generation, thus demanding a more rigorous approach to dataset curation and algorithmic fairness controls.

This concern warrants a more critical approach to AI-assisted creation, one that explicitly considers dataset constraints, sustainability, and authorship, instead of treating biased outputs as merely isolated technical failures [2].

2.4. Forensic Transparency and the Fight Against Manipulation

The rise of the `hyper-realistic generation` has made the distinction between reality and simulation increasingly blurred [7].Peregrina argues that the majority of users cannot reliably distinguish between AI-generated and human-created images, leading to social media communications and national security vulnerabilities [7].

To counter the threats by deepfakes and uncompensated appropriation, new frameworks for `ethical attribution` are emerging. The Compositional Provenance Framework (CPF) proposed by Sankaran uses `retrieval-augmented influence tracking` and `cryptographic governance` to log attribution metadata directly into the generative pipeline [6].

This technical solution aim to make creative borrowing visible and contestable, thereby providing a path toward the ethical exploitation of generative models.

3. FUTURE DIRECTIONS

Current research is moving toward two fronts: `Edge AI` and `Opt-in Training Models`. Future work should prioritize real-time physical consistency, decentralized attribution, and cross-modal fairness.

Note

The training corpora of **Opt-in Training Models** are composed exclusively of assets contributed under explicit, revocable consent, with license, and provenance bound to each asset and enforced across ingestion, training, release, and post-training takedown.

Table 1. Prioritized Roadmap

Priority	Tasks
Real-time Physical Consistency	Refining the integration of continuum mechanics in generative video to avoid visual artifacts in complex game environments.
Decentralized Attribution	Leveraging blockchain and the CPF framework to ensure artists are compensated when their style is utilized by commercial models.
Cross-Modal Fairness	Developing algorithms that actively de-bias stereotypes during the inference phase, ensuring diverse and respectful representation in audiovisual content.

4. CONCLUSION

The systematic review of recent literature reveals a twofold reality, while deep-learning offers unprecedented acceleration for audiovisual production and 3D modeling, it often operates within an ethical vacuum regarding copyright and social bias.

The transition from `black-box` model to transparent, `provenance-aware systems` is not merely just a technical requirement but a moral imperative. As the boundaries of reality continue to shift, the industry must adopt a human-in-the-loop philosophy where AI serves the artist's vision without eroding the foundational rights of human creators.

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